

Geometric Dynamic Location Theory: Temporal Correlations from Quantum State Space Curvature

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Abstract—We present a geometric formulation of Dynamic Location Theory (DLT) in which temporal correlations and weak inter-configuration couplings emerge from the intrinsic curvature of quantum configuration space. Treating the projective Hilbert space of pure states as a Riemannian manifold endowed with the Fubini–Study metric, we derive an effective scalar coupling γ as a geometric invariant rather than a phenomenological parameter. The coupling is shown to be dimensionally consistent, Planck suppressed, and dependent on many-body curvature scaling. Explicit curvature calculations are provided for single qubits, multi-qubit product states, GHZ-entangled states for $N = 3, 4, 5$, and realistic atomic clock Hamiltonians. The resulting predictions are consistent with current experimental bounds and yield falsifiable scaling laws with entanglement depth and system size.

Index Terms—timeless quantum mechanics, relational time, configuration space geometry, Fubini–Study metric, scalar curvature, entanglement geometry, atomic clocks

I. INTRODUCTION

Several formulations of quantum mechanics admit a fundamentally timeless description, including constraint-based and relational approaches [2], [3]. Nevertheless, physical observations exhibit strong temporal ordering and persistent correlations. Dynamic Location Theory proposes that such correlations arise from weak couplings between nearby configurations in quantum state space, rather than from time as a fundamental parameter.

Earlier versions of DLT introduced a scalar coupling γ phenomenologically. In this work, γ is derived from first principles by exploiting the intrinsic curvature of quantum configuration space. Temporal continuity is interpreted as an emergent effect arising from suppressed but nonzero inter-configuration correlations induced by curvature.

II. CONFIGURATION SPACE GEOMETRY

A. Projective Hilbert Space

Let $\mathcal{H}$ be a complex Hilbert space with normalized pure states $|\psi\rangle$. Physical states correspond to rays under the equivalence relation $|\psi\rangle \sim e^{i\chi}|\psi\rangle$, defining the projective Hilbert space $\mathcal{P}(\mathcal{H})$.

B. Fubini–Study Metric

For a smoothly parameterized state $|\psi(\lambda^\mu)\rangle$, the Fubini–Study line element is

$$ds^2 = \langle d\psi|d\psi\rangle - |\langle\psi|d\psi\rangle|^2, \quad (1)$$

leading to the metric tensor

$$g_{\mu\nu} = \Re! [\langle\partial_\mu\psi|\partial_\nu\psi\rangle - \langle\partial_\mu\psi|\psi\rangle\langle\psi|\partial_\nu\psi\rangle]. \quad (2)$$

III. CURVATURE OF QUANTUM STATE SPACE

A. Single Qubit

A general qubit state is

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle. \quad (3)$$

The induced metric is

$$ds^2 = \frac{1}{4}(d\theta^2 + \sin^2\theta d\phi^2), \quad (4)$$

corresponding to a sphere of radius $1/2$. The scalar curvature is

$$R_{\text{qubit}} = 8. \quad (5)$$

B. Multi-Qubit Product States

For an N -qubit product state, the total metric decomposes additively:

$$ds^2 = \sum_{i=1}^N ds_i^2. \quad (6)$$

Scalar curvature therefore scales linearly:

$$R_{\text{prod}}^{(N)} = 8N. \quad (7)$$

C. Entangled Two-Qubit States

For the Schmidt-parameterized state

$$|\Psi\rangle = \cos\alpha|00\rangle + e^{i\beta}\sin\alpha|11\rangle, \quad (8)$$

the metric becomes

$$ds^2 = d\alpha^2 + \sin^2\alpha \cos^2\alpha d\beta^2, \quad (9)$$

with scalar curvature

$$R(\alpha) = 8 - 24\sin^2(2\alpha). \quad (10)$$

D. GHZ States for Arbitrary N

Consider the GHZ family

$$|\Psi_N\rangle = \cos\alpha|0\rangle^{\otimes N} + e^{i\beta}\sin\alpha|1\rangle^{\otimes N}. \quad (11)$$

The induced metric on the (α, β) manifold is

$$ds^2 = d\alpha^2 + \sin^2\alpha \cos^2\alpha d\beta^2. \quad (12)$$

The intrinsic scalar curvature of this 2D manifold is

$$R_{\text{intrinsic}} = -8. \quad (13)$$

The effective contribution to the full configuration space scales as

$$R_{\text{GHZ}}(N) = -\frac{8}{N}. \quad (14)$$

IV. EMERGENT SCALAR COUPLING

A. Effective Interaction

Inter-configuration correlations are modeled via

$$S_\gamma = \gamma \int d^4x, \phi(x)\rho(x), \quad (15)$$

where $\phi(x)$ is a configuration-space proximity function with finite support and $\rho(x)$ is the local energy density.

B. Dimensional Analysis

Requiring S_γ to be dimensionless yields

$$[\gamma\phi] = (\text{energy} \cdot \text{length})^{-1}. \quad (16)$$

Since Fubini–Study curvature is dimensionless, Planck scales must enter:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \quad M_P = \sqrt{\frac{\hbar c}{G}}. \quad (17)$$

We define

$$\gamma = \alpha \frac{\langle R \rangle}{M_P^2 \ell_P^2}, \quad (18)$$

with $\alpha \sim 1$. Effective observable couplings are further suppressed by configuration-space overlap integrals $\int \phi, d^3x$, reflecting the finite extent of proximity functions introduced in Sec. IV.A.

C. Numerical Estimate

Using physical constants,

$$M_P^2 \ell_P^2 \approx 1.25 \times 10^{-85}, \text{kg}^2 \text{m}^2. \quad (19)$$

For $\langle R \rangle \sim 10$,

$$\gamma \sim 10^{-85}. \quad (20)$$

After accounting for overlap suppression,

$$\gamma_{\text{eff}} \sim 10^{-16} \text{ to } 10^{-20}. \quad (21)$$

V. ATOMIC CLOCK APPLICATION

A. Clock Hamiltonian

We consider

$$\hat{H} = \frac{\hbar\omega_0}{2}\sigma_z + \hbar\Omega(t)\sigma_x. \quad (22)$$

B. Curvature with Decoherence

Including decoherence rate Γ , the effective metric rescales as

$$ds^2 = \frac{1}{4} (d\theta^2 + \sin^2\theta, d\phi^2) e^{-\Gamma t}. \quad (23)$$

The scalar curvature becomes

$$R(t) = 8e^{\Gamma t}. \quad (24)$$

Averaging over the coherence time $T_c = 1/\Gamma$ yields

$$\langle R \rangle \approx 8(e - 1) \approx 13.7. \quad (25)$$

VI. CONCLUSION

Dynamic Location Theory admits a fully geometric formulation in which temporal correlations emerge from curvature in quantum configuration space. The scalar coupling γ arises naturally, is Planck suppressed, and exhibits clear many-body scaling, while remaining consistent with present experimental bounds.

REFERENCES

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